Task-Dependent Optimal Linear Observers in Linear Gaussian Dynamical Systems

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For a stationary linear Gaussian dynamical system, the rank-r linear observer minimizing squared-error prediction loss for target $Cx_{t+\tau}$ has row space equal to the top-r eigenspace of the target information matrix $M_{C,\tau} = \Sigma^{1/2}(A^\tau)^\top C^\top C A^\tau \Sigma^{1/2}$, uniquely up to eigenspace rotation. Two targets with distinct dominant eigenspaces admit no common optimal rank-r observer. A common optimal observer exists if and only if the target family is r-*coherent*: all information matrices share a common top-r eigenspace. This condition is exact, not asymptotic or approximate. Numerical illustrations at $n = 8$, $r = 2$ confirm a target-dependent case ($\Gamma = 0.026 > 0$) and an r-coherent construction, both consistent with the theorem.

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I. INTRODUCTION

A central question in observer design for linear dynamical systems is whether a single rank-r linear observer can be simultaneously optimal for predicting multiple output channels $\{C_k x_{t+\tau_k}\}$, or whether the optimal observer depends on the specific prediction target and horizon. For finite-dimensional linear Gaussian dynamical systems (LGDS) at stationarity, this paper gives an exact answer: the optimal rank-r observer for predicting $Cx_{t+\tau}$ is the top-r eigenspace of the target information matrix $M_{C,\tau}$, and a common optimal observer across a family of targets exists if and only if all information matrices share a common dominant eigenspace.

This paper is not about dynamic observer synthesis in the Luenberger–Kalman sense [1–3]: the observer is a fixed rank-r linear map $\phi_U(x) = U\Sigma^{-1/2}x$, and performance is squared-error prediction loss for a specified target $Cx_{t+\tau}$ at stationarity. The static case ($A = I$, $\tau = 0$) recovers standard principal-subspace selection, but the main object here is the interaction of system dynamics, prediction horizon τ, and observer rank through the information matrix $M_{C,\tau}$.

Reduced representations for linear systems have been studied via principal components of Gramians [4], balanced truncation, and related model-reduction methods [5]. The observability Gramian accumulates contributions from all prediction horizons and all output directions; balanced truncation minimises approximation error in H_2/H_∞ system norms rather than squared-error prediction loss for a specified (C, τ). Sensor-selection and output-design methods target estimation or reconstruction quality for a given objective, typically without characterising when a fixed low-rank map is simultaneously optimal across an entire task family. The present note asks precisely that structural question: given distinct prediction targets with information matrices $\{M_{C_k,\tau_k}\}$, when is one rank-r observer simultaneously optimal for all of them, and when is target-dependent selection unavoidable?

The Grassmannian $\mathrm{Gr}(r, n)$ is the natural parameter space for rank-r linear observers [6], and observer quality—the trace $\mathrm{tr}(UM_{C,\tau}U^\top)$—is a smooth function on this manifold. Optimal observer selection reduces to trace maximisation on $\mathrm{Gr}(r, n)$, governed by the Ky Fan variational principle [7]: the maximum is achieved by the leading eigenspace of $M_{C,\tau}$, recovering principal-subspace selection [8] in the static ($A = I$) case within the classical linear state-space framework [3].

The Ky Fan variational principle and principal-subspace optimality are classical [7, 8]. The single-target result (Proposition 2) applies Ky Fan to the target information matrix $M_{C,\tau}$, which identifies which directions of state space bear on a given prediction task. What is not resolved by classical eigenspace theory is the task-family question: when does a single rank-r observer remain exactly optimal across a family of targets with possibly distinct information matrices? The structural characterization, given in Corollary 6, is r-coherence: its non-trivial content is the backward direction, which shows that any commonly optimal observer must be the unique top-r eigenspace of each individual M_k, forcing all information matrices to share a common dominant eigenspace. The regret gap Γ provides a computable certificate of unavoidable suboptimality when r-coherence fails.

The result is scoped to finite-dimensional LGDS, fixed-rank linear observers, and squared-error prediction targets.

II. SYSTEM AND OBSERVER FAMILY

a. Dynamics. Let

$$x_{t+1} = Ax_t + \xi_t, \quad \xi_t \sim \mathcal{N}(0, Q), \quad x_t \in \mathbb{R}^n, \tag{1}$$

with A stable (spectral radius < 1) and $Q \succ 0$. The unique stationary covariance Σ satisfies $A\Sigma A^\top + Q = \Sigma$ and is strictly positive definite [3]; $\Sigma^{1/2}$ denotes its symmetric positive-definite square root.

b. Observer family. An observer of rank r ($1 \le r < n$) is parametrised by a row-orthonormal matrix $U \in \mathbb{R}^{r\times n}$, $UU^\top = I_r$:

$$\phi_U(x) = U\Sigma^{-1/2}x. \tag{2}$$

Pre-whitening by $\Sigma^{-1/2}$ places all observers on a common isotropic footing; in the static case ($A = I$, $\tau = 0$) this reduces to principal component analysis [8]. Two observers differing only by an orthogonal rotation of their row spaces are equivalent; the observer space is the Grassmannian $\mathrm{Gr}(r, n)$, a smooth compact Riemannian manifold of dimension $r(n-r)$, with metric and geodesic structure from $O(n)$ [6].

c. Target family. A prediction target is a pair (C, τ) with $C \in \mathbb{R}^{p\times n}$ and integer $\tau \ge 1$:

$$y_{C,\tau} = Cx_{t+\tau}. \tag{3}$$

Performance is total mean squared prediction error. A target family $\mathcal{T} = \{(C_k, \tau_k)\}$ is any finite collection.

III. THEOREM

A. Exact Bayes risk formula

Given observation $z = U\Sigma^{-1/2}x_t$, the MMSE predictor of $Cx_{t+\tau}$ is linear by Gaussian conjugacy. At stationarity, $x_{t+\tau} = A^\tau x_t + w_\tau$ where w_τ is the accumulated process noise independent of x_t, so $E[x_{t+\tau}x_t^\top] = A^\tau\Sigma$. It follows that $\mathrm{Cov}(Cx_{t+\tau}, z^\top) = CA^\tau\Sigma \cdot \Sigma^{-1/2}U^\top = CA^\tau\Sigma^{1/2}U^\top$ and $\mathrm{Cov}(z, z^\top) = U\Sigma^{-1/2}\Sigma\Sigma^{-1/2}U^\top = I_r$. Applying the Gaussian linear MMSE formula $\mathcal{R}(U; C, \tau) = \mathrm{tr}(\mathrm{Var}(Cx_{t+\tau})) - \mathrm{tr}\big(\mathrm{Cov}(Cx_{t+\tau}, z^\top)\mathrm{Cov}(z, z^\top)^{-1}\mathrm{Cov}(z, Cx_{t+\tau})\big)$ gives

$$\mathcal{R}(U; C, \tau) = \mathrm{tr}(C\Sigma C^\top) - \mathrm{tr}\big(U\, M_{C,\tau}\, U^\top\big), \tag{4}$$

where the *target information matrix* is

$$M_{C,\tau} = \Sigma^{1/2}(A^\tau)^\top C^\top CA^\tau\Sigma^{1/2} \in \mathbb{R}^{n\times n}. \tag{5}$$

$M_{C,\tau}$ is symmetric positive semidefinite of rank at most $\min(p, n)$. The formula is exact.

Remark 1. *In the static case ($A = I$, $\tau = 0$), $M_{C,0} = \Sigma^{1/2}C^\top C\Sigma^{1/2}$, which differs from the state covariance Σ used by PCA unless $C = I_n$. For $C = e_k^\top$ (predicting one component), $M_{C,0}$ is rank-1 with eigenvector e_k in the whitened basis: the optimal rank-1 observer captures the k-th whitened direction regardless of the state variance in other components, whereas PCA selects the maximum-variance direction. In the dynamic case, the rotation by $(A^\tau)^\top$ in $M_{C,\tau}$ shifts the dominant eigenspace with τ: the same target C admits a horizon-dependent optimal observer.*

B. Top-eigenspace optimality

Proposition 2. *For fixed rank r, the observer minimising $\mathcal{R}(U; C, \tau)$ over $\mathrm{Gr}(r, n)$ has row space equal to the top-r eigenspace of $M_{C,\tau}$. The minimum risk is*

$$\mathcal{R}^* = \mathrm{tr}(C\Sigma C^\top) - \sum_{i=1}^{r}\lambda_i(M_{C,\tau}), \tag{6}$$

where $\lambda_1 \ge \cdots \ge \lambda_n$ are the eigenvalues of $M_{C,\tau}$. Uniqueness up to rotation holds when $\lambda_r > \lambda_{r+1}$.

Degeneracy at the gap ($\lambda_r = \lambda_{r+1}$) yields a continuum of optimal observers spanning the degenerate eigenspace; in practice, a generic target (C, τ) produces a strict spectral gap, making degeneracy non-generic for fixed (A, Σ).

Proof. Minimising $\mathcal{R}$ is equivalent to maximising $\mathrm{tr}(UM_{C,\tau}U^\top)$ subject to $UU^\top = I_r$. By the Ky Fan variational principle [7, 9] the maximum equals $\sum_{i=1}^r \lambda_i(M_{C,\tau})$, achieved exactly when the row space of U spans the corresponding eigenspace. ■

The *task regret* of observer U for target (C, τ) is the excess Bayes risk:

$$\mathrm{Reg}(U; C, \tau) := \mathcal{R}(U; C, \tau) - \mathcal{R}^*(C, \tau) = \sum_{i=1}^{r}\lambda_i(M_{C,\tau}) - \mathrm{tr}(U\, M_{C,\tau} \tag{7}$$

Corollary 3 (Regret–angle bound)**.** *Let $M = M_{C,\tau}$ with eigenvalues $\lambda_1 \ge \cdots \ge \lambda_n \ge 0$ and top-r eigenspace U^*. For any $U \in \mathrm{Gr}(r, n)$:*

$$(\lambda_r - \lambda_{r+1})\,\|\sin\Theta\|_F^2 \le \mathrm{Reg}(U; C, \tau) \le (\lambda_1 - \lambda_n)\,\|\sin\Theta\|_F^2, \tag{8}$$

where $\|\sin\Theta\|_F^2 := \|\sin\Theta(U, U^*)\|_F^2$ [10] is the squared Frobenius norm of the principal-angle sines between $\mathrm{row}(U)$ and $\mathrm{row}(U^*)$.

Proof. Let $p_i := \sum_{k=1}^r (v_i^\top u_k)^2$ where v_i are eigenvectors of M and u_k are orthonormal rows of U. Then $\mathrm{Reg}(U) = \sum_{i\le r}\lambda_i(1 - p_i) - \sum_{i>r}\lambda_i p_i$ and $\|\sin\Theta\|_F^2 = \sum_{i\le r}(1 - p_i) = \sum_{i>r} p_i$ [10]. Centering at λ_{r+1}:

$$\mathrm{Reg}(U) = \underbrace{\textstyle\sum_{i\le r}(\lambda_i - \lambda_{r+1})(1 - p_i)}_{\ge 0} + \underbrace{\textstyle\sum_{i>r}(\lambda_{r+1} - \lambda_i)\, p_i}_{\ge 0}.$$

Lower bound: drop the second sum; bound $\lambda_i - \lambda_{r+1} \ge \lambda_r - \lambda_{r+1}$ for $i \le r$. *Upper bound*: bound the first coefficient by $\lambda_1 - \lambda_{r+1}$ and the second by $\lambda_{r+1} - \lambda_n$; the two $\|\sin\Theta\|_F^2$ factors sum as written. ■

C. Target-dependent observer selection

Corollary 4. *If targets (C_1, τ_1) and (C_2, τ_2) have information matrices M_1 and M_2 with distinct top-r eigenspaces, no single rank-r observer is simultaneously optimal for both.*

Proof. Optimality for target k requires the row space of U to equal the top-r eigenspace of M_k. Distinct eigenspaces cannot be spanned by the same U. ■

The corollary extends to any finite target family: if $\{M_k\}$ do not share a common top-r eigenspace, no common optimal rank-r observer exists.

D. r-coherence: the exact exception

Definition 5. *A target family $\mathcal{T}$ is* r-coherent *with respect to (A, Σ) if all matrices $\{M_{C,\tau} : (C, \tau) \in \mathcal{T}\}$ share a common top-r eigenspace V^*.*

Corollary 6. *A common optimal observer exists if and only if $\mathcal{T}$ is r-coherent.*

Proof. (*If*) If $\mathcal{T}$ is r-coherent with shared top-r eigenspace V^*, the observer with row space V^* minimises $\mathcal{R}(\cdot; C_k, \tau_k)$ for each k by Proposition 2. (*Only if*) If U is simultaneously optimal for all k, then by Proposition 2's uniqueness (holding when $\lambda_r(M_k) > \lambda_{r+1}(M_k)$ for all k), the row space of U must equal the top-r eigenspace of every M_k; hence all M_k share that eigenspace and $\mathcal{T}$ is r-coherent. ■

r-coherence requires all targets to be functions of a common r-dimensional sufficient statistic across all prediction horizons—a strong alignment between the system dynamics and the task family. Outside r-coherence, the optimal observer is target-dependent.

Remark 7. *r-coherence is an algebraic equality between eigenspaces, not an approximate or asymptotic condition. It is the exact condition for a common optimal observer to exist.*

Remark 8. *r-coherence is strictly weaker than simultaneous diagonalisability of $\{M_k\}$: only the dominant r-dimensional eigenspace must coincide, not the full eigenbasis. Simultaneous diagonalisability (equivalently, pairwise commutativity $[M_i, M_j] = 0$) implies r-coherence for every r, but not conversely. For generic (A, Σ) and target families, the top-r eigenspaces of distinct M_k are distinct, so r-coherence is a non-generic condition absent structural alignment between the task family and the system.*

E. Minimax shared observer and approximate coherence

For a task family $\mathcal{T} = \{(C_a, \tau_a)\}_{a=1}^m$, define the *minimax shared observer* and *minimax regret gap*:

$$U_{\mathrm{mm}} = \arg \min_{U \in \mathrm{Gr}(r,n)} \max_a \mathrm{Reg}(U; C_a, \tau_a), \qquad \Gamma := \min_{U \in \mathrm{Gr}(r,n)} \max_a \mathrm{Reg}(U; C_a, \tau_a). \tag{9}$$

U_{mm} is the best shared rank-r observer in the worst-case sense; $\Gamma \geq 0$ is the unavoidable worst-case cost of any single observer.

Corollary 9. $\Gamma = 0$ *if and only if $\mathcal{T}$ is r-coherent.*

Proof. (*If*) If r-coherent with shared eigenspace V^*: $\mathrm{Reg}(V^*; C_a, \tau_a) = 0$ for all a by Proposition 2, so $\Gamma = 0$. (*Only if*) If $\Gamma = 0$, some U_{mm} achieves $\mathrm{Reg}(U_{\mathrm{mm}}; C_a, \tau_a) = 0$ for all a, so U_{mm} is optimal for every task; the only-if direction of Corollary 6 then gives r-coherence. ■

Corollary 10 (Approximate coherence)**.** *If there exists $U \in \mathrm{Gr}(r, n)$ with $\|\sin \Theta(U, U_a^*)\|_F^2 \leq \varepsilon^2$ for all a, then $\Gamma \leq \Lambda\, \varepsilon^2$, where $\Lambda := \max_a[\lambda_1(M_a) - \lambda_n(M_a)]$.*

Proof. Apply the upper bound of Corollary 3 to U for each task and take the maximum over a. ■

IV. NUMERICAL ILLUSTRATIONS

The theorem is analytically exact under its stated assumptions. We illustrate it numerically with two finite-dimensional instances: a target-dependent case and an r-coherent case. All reported quantities are computed exactly from the analytic Bayes risk formula and numerical optimization on the Grassmannian.

a. Setup. State dimension $n = 8$, observer rank $r = 2$. The model is a finite-dimensional linear Gaussian dynamical system [11]. $A = VDV^{-1}$ with eigenvalues drawn from $U[0.3, 0.85]$, random non-orthogonal V (seed 100), $Q = 0.1\, I_8$. Σ is computed from the discrete Lyapunov equation; $\Sigma \succ 0$ is confirmed to machine precision. All reported quantities are computed exactly from the analytic Bayes risk formula, eigenspace calculations, and repeated projected optimization over $\mathrm{Gr}(2, 8)$ (300 random initializations of projected L-BFGS-B, gradient-norm threshold 10^{-12}, function-value tolerance 10^{-15}).

b. Condition A: target-dependent instance. Targets $T_1 = ([e_1^\top; e_2^\top], 1)$ and $T_2 = ([e_5^\top; e_6^\top], 5)$.

With non-normal A, A^1 and A^5 rotate state-space directions differently; the principal angles [10] between the top-2 eigenspaces of M_1 and M_2 are 54.1° and 24.0°, confirming that no single rank-2 observer is simultaneously optimal.

The task regret (7) and minimax gap Γ (9) are computed from the exact Bayes risk formula. By Corollary 9, $\Gamma > 0$ certifies non-r-coherence. The numerically computed minimax observer U_{joint} achieves $\Gamma = 0.026 > 0$; balanced regrets on both targets are a necessary condition for a minimax solution, consistent with the global minimum having been attained. Table I gives the full regret structure (7).

c. Condition B: structurally coherent example. A target family is constructed so that all M_k share the same top-2 eigenspace by design: normal $A_{\mathrm{sym}} = VDV^\top$ with orthogonal V, and a common C whose rows are the top-2 eigenvectors of A, varying only the prediction horizon $\tau \in \{1, 2, 3, 5\}$. Under this construction $M_{C,\tau}$ has the same dominant 2-dimensional eigenspace for all τ, so the four optimal observers are identical by the theorem. The

TABLE I. Regret matrix, Condition A ($n = 8$, $r = 2$, seed 100). All values from the exact analytic Bayes risk formula.

Observer	Regret on T_1	Regret on T_2
U_1^* (optimal for T_1)	0.000	0.036
U_2^* (optimal for T_2)	0.763	0.000
U_{joint} (compromise)	0.026	0.026

maximum pairwise principal angle across all four is 0.000° to machine precision.

Under $\varepsilon = 0.1$ perturbations to C (20 random draws), the maximum principal angle across all draws and all pairs is 0.102 rad—well below the fragility threshold of 0.15 rad, set at 1.5ε. The threshold 1.5ε is a heuristic calibration; the direct numerical check confirms that the shared dominant eigenspace is robust to this level of perturbation in C. Structural coherence is perturbation-robust, distinguishing it from accidental near-coincidences of eigenspaces that break under small perturbations.

The numerical results are consistent with the theorem in all cases.

V. SCOPE

a. What is earned. In stationary finite-dimensional LGDS, among rank-r linear observers $\phi_U(x) = U\Sigma^{-1/2}x$ and squared-error prediction targets $y = Cx_{t+\tau}$, no common optimal rank-r observer exists unless the target family is r-coherent. r-coherence requires all prediction targets to be functions of a common r-dimensional sufficient statistic across all horizons—a strong alignment between the system dynamics and the task family. Outside r-coherence, no common optimal observer exists, and the cost of applying any fixed observer to a non-native target is given exactly by the Bayes risk formula. In observer design workflows where a single rank-r map must serve multiple prediction channels, Γ is therefore a computable lower bound on worst-case suboptimality under any shared observer: if $\Gamma > 0$, dedicated per-target observers are necessary to achieve optimality. The Grassmannian $\mathrm{Gr}(r, n)$ is the literal domain of this optimisation: a compact Riemannian manifold on which observer quality is a smooth function, with $O(n)$ acting by transformation. For fixed C, the computation reduces to a single $n \times n$ eigendecomposition of $M_{C,\tau}$ per horizon once A^τ is formed; A^τ is obtained in $O(n^3 \log \tau)$ operations by repeated squaring.

b. What is not earned. This result concerns optimal reduced linear observation for prediction; it does not address dynamic observer synthesis, asymptotic state reconstruction, or feedback design. It also does not apply to nonlinear systems, non-Gaussian noise, nonlinear observers, or non-prediction tasks. The breakdown is structural: for non-Gaussian noise or nonlinear dynamics, the MMSE predictor is generally nonlinear and Proposition 2's trace maximisation has no closed-form counterpart; for nonlinear observers, the Grassmannian parametrisation and the Ky Fan principle both cease to apply. The causal axis—whether the observer optimal for intervention differs from the observer optimal for prediction—is not addressed here.

c. Extensions. The linear Gaussian setting admits closed-form analysis because the MMSE predictor is linear, the stationary distribution is Gaussian, and the optimal observer reduces to a single eigendecomposition. Whether analogous shared-eigenspace conditions characterise common optimal observers in non-Gaussian or nonlinear systems, and whether intervention objectives impose additional constraints on observer selection, are open questions beyond the scope of the present result.

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